Risk in a Shared Public–Private Pool

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The Questions

Our main fund has both privates and publics in a shared pool (not side-pocketed). How should we conceptually think about risk given this?

- Should we manage risk in the public book to overall NAV, or just public NAV, or some mixture? How does this vary with different % private, and different timelines to liquidity?
- What are the risk considerations we should take into account when deciding on a marginal private investment? How does this vary with different % private?
    - How should we think about the risk and liquidity considerations between privates that are close to IPO and likely to succeed vs. more call-option-y earlier stage privates?
    - What does that imply for the IRR threshold for a private vs. a public (taking into account cost of liquidity, cost of risk, and leverage constraints)?

Simplified constraints/setting:

- Capital is locked up for a long time, and if necessary we can put things in "liquidating capital accounts" (i.e. redeem in kind rather than in cash).
    - So there can't be an inability to pay / run on the bank issue.
    - That said, broadly the expectation is that folks will be able to redeem in cash in normal circumstances (where there is only a small percentage redeeming at a given time), so it's costly for us if we end up with a very high % illiquid.
        - High illiquidity (say, >50%) introduces further risks such as valuation and potentially broader investor frustration
- People see and judge an overall performance number on overall NAV, so most of the cost of drawdowns from an IR etc. perspective is on the overall NAV. (Below some limit, such as the <50% above)
- We've generally guided to 20–30% private, as target allocations, but not hard constraints.
- Representatively, consider:
    - **Private A** (mega cap private): Very likely to go public in the next 12 months, like a large cap stock. (Ex.: Anthropic)
    - **Private B** (late stage private): Likely to go public in the next 12–36 months, more risky than A but successful in most worlds. (Ex: Fluidstack)
    - **Private C** (early stage private): Isn't successful in 70% of worlds, in 30% of worlds is wildly successful (a bit like a call option). Will take a few years to prove out. (Ex.: Source Foundry)
- For thinking about how to think about the risk of the public sleeve of the book, consider
    - **Setting 1:** Fund is 15% private (10% Private B, 5% Private C)
    - **Setting 2:** Fund is 60% private (45% Private A, 10% Private B, 5% Private C).

        - In this setting, how much more volatility (relative to public NAV) can we take on the currently-public sleeve of the book? How close to managing to a denominator of overall NAV vs. just the public sleeve should we be?
    - Broadly think of bottom line utility of the fund as just log($) in the long run
    - We can fundraise (bringing in additional liquid $ reducing % private)
        - Though fundraising is harder when you've lost a lot of money and/or privates are extremely high.

Measure and size risk on total economic NAV, and treat illiquidity as two separate things: a smaller effective denominator for the public book's alpha, and a cash constraint along adverse paths. At 15% private, the public book should be managed to total NAV (about 1.13× the vol an all-public fund would run). At 60% private, the public book should first hedge the privates' theme exposure; its alpha can then run about 1.3–2.2× the all-public level, and a marginal private needs roughly 5–30 more points of expected return than at 15%.

All numbers below come from stipulated profiles for A, B and C and from stated parameters. They show the mechanism and the orders of magnitude; they are not forecasts about the named companies.

Short answers

Which denominator for the public book? It depends on the kind of risk.

- **Theme and factor exposure:** always total economic NAV. The public book is the fund's *completion portfolio*: it holds whatever exposure brings the whole fund to its target, net of what the privates already carry.
- **Idiosyncratic alpha:** an effective denominator $D = L + \theta \times P$, where θ is the credit given to private NAV (between 0 and 1). θ falls as the private share rises, as the privates get more volatile, and as time to usable liquidity lengthens (Steps 3–4).

Setting 1 (15% private). $\theta \approx 0.7$–0.8. The public book can run about 1.13× the vol of an all-public fund; full credit would give 1.18×. In practice, manage to total NAV.

Setting 2 (60% private). Before sizing alpha, the public book should be net short the theme by about 25% of NAV. After that, θ is 0.2–0.8 (central about 0.45), and the alpha book can run about 1.3–2.2× the all-public vol (central 1.7×), or 1.0–2.0× if A's exit slips. Full credit would give 2.5×. Relative to Setting 1, that is about 1.5×, not the mechanical 2.125×. A 40% liquid sleeve cannot carry that hedge with futures or swaps through a strong rally, so it should use puts, a smaller hedge, or a smaller A (Step 5).

What to weigh on a marginal private. Once the public book re-hedges the private's theme beta, the private must beat its theme-implied return by an alpha that covers four costs:

1. the capital it pulls from the public book;
2. the concentration it adds;

3. its own ticket size;

4. the commercial cost of raising the private share.

Concentration and the commercial cost both rise with the private share. The commercial cost is zero below about 30% and steep above 50% (Step 6).

Near-IPO vs call-option privates.

- **A (near-IPO):** mostly hedgeable market risk; its danger is size.
- **B (late stage):** the most dangerous per dollar, because its liquidity date slips in exactly the bad states.
- **C (early stage):** cheap in risk terms if small and spread across names; its costs are skew, follow-ons, and correlation with the same boom (Step 7).

IRR thresholds. These are expected (probability-weighted) annual returns (Step 8).

- **At 15% private:** A needs its theme-implied return plus about 0–9 points, and B plus about 4–13. The low end applies when the public book is capacity-bound, the high end when it is capital-bound.
- **At 60% private:** add roughly 5–30 more points.
- **C:** needs an expected-value IRR of about 4–16% at 15% private and 9–29% at 60%. The stipulated C offers 24.5%.

Setup and definitions

Balance sheet. Normalize total NAV to 1:

$$N = L + P, \qquad \pi = \frac{P}{N}$$

- L **(liquid sleeve):** the net value of the public book: cash, longs, shorts and hedges.
- P **(private holdings):** measured at economic value, not at their marks.
- π **(private share):** the fraction of NAV in privates.

Three values of a private. Keep these apart; most confusion comes from mixing them.

Concept	Definition	Used for
Economic value	Expected value of the payoff given today's information; moves with public comparables	Risk, sizing, the objective
Reported mark	The fund's valuation; lags and smooths economic value	Reported NAV, subscription and redemption prices, what investors see
Usable liquidity	Cash obtainable within a horizon, after lockups, block size and discounts	Margin, redemptions, follow-ons, expenses

Time to usable liquidity (T). The date by which a meaningful part of the position can actually be sold. It is not the IPO date: for a large block it includes the lockup and a sell-down period. Each private gets a base value and a stress value, because exits slip in bad markets.

Objective. Maximize $\mathbb{E}[\log W]$, where W is NAV per share of a continuing investor. Per share matters: new subscriptions add liquid capital, but they are not returns to existing investors.

Log growth. For a portfolio with expected excess return μ and variance σ^2, the expected log growth rate is approximately:

$$g \approx \mu - \tfrac{1}{2}\sigma^2$$

Kelly sizing. For a single bet, maximizing g gives the log-optimal fraction of wealth:

$$w^\star = \frac{\mu}{\sigma^2}$$

Fractional Kelly. Scale the Kelly fraction by κ. $\kappa = \frac{1}{2}$ is standard because μ is estimated with error; it is equivalent to constant relative risk aversion $\gamma = 1/\kappa$.

Theme factor and betas. F is the AI/tech theme, tradeable through public baskets. Each private j has:

- correlation ρ_j with F and volatility σ_j;
- beta $\beta_j = \rho_j \sigma_j / \sigma_F$;
- idiosyncratic variance $\sigma_{\varepsilon j}^2 = \sigma_j^2(1 - \rho_j^2)$;
- alpha α_j: its expected return beyond the theme-implied return $\beta_j \times \mu_F$.

The two settings.

Holding (% of NAV)	Setting 1	Setting 2
Liquid sleeve L	85	40
A (near-IPO mega-cap)	0	45
B (late stage)	10	10
C (early stage)	5	5

Parameters. The A/B/C profiles are stipulated from the prompt, not forecasts about the named companies.

Item	Value used	Notes
Theme F	vol 35%/yr, expected excess return 6%/yr	Only used to set the target theme exposure
A	vol 45%, correlation 0.7 → beta 0.90, idiosyncratic vol 32%	Usable liquidity 2.5y base (≤1y to IPO, 6-month lockup, ∼1y block sell-down), 4y stress
B	vol 65%, correlation 0.5 → beta 0.93, idiosyncratic vol 56%	Usable liquidity 3y base, 5y stress
C	30%: 8× in 4 years; 70%: zero	Expected multiple 2.4×; expected-value IRR 24.5%; IRR if it works 68%
Public alpha book	12%/yr excess return on capital; vol 10% of public NAV in an all-public fund; uncorrelated with the privates	The baseline every multiplier is quoted against
Margin	20% of notional on public longs or shorts; 100% on privates	A private has zero collateral value
Commercial cost of a high private share	f_0 per year at 60% private, starting at 30%; $f_0 = 1\%$ or 3%	Step 4
Redemptions, follow-ons, expenses	5%, 2% and 1% of NAV per year in normal times	Step 5

Step 1 — Measure the fund's risk on economic NAV

In Setting 2 the privates alone carry about 23% of NAV in annual volatility, yet reported NAV would show roughly half the fund's true vol. The risk model therefore has to map privates to public comparables, not to their marks.

Factor model. Each private's return is a theme component plus its own risk:

$$r_j = \beta_j F + \varepsilon_j, \qquad \beta_j = \frac{\rho_j \sigma_j}{\sigma_F}, \qquad \mathrm{Var}(\varepsilon_j) = \sigma_j^2 (1 - \rho_j^2)$$

With weights w_j (fractions of NAV), the privates' theme dollar-beta is $B = \sum_j w_j \beta_j$. If the idiosyncratic terms are uncorrelated, the privates' contribution to NAV variance is:

$$\sigma^2_{\text{priv}} = B^2 \sigma_F^2 + \sum_j w_j^2 \, \sigma^2_{\varepsilon j}$$

Worked numbers.

Contribution to NAV vol	Setting 1	Setting 2
Theme dollar-beta B	$0.10 \times 0.93 = 0.093$	$0.45 \times 0.90 + 0.10 \times 0.93 = 0.498$
Theme vol, $B \times 35\%$	3.3%	17.4%
A's own risk, $0.45 \times 32.1\%$	—	14.5%
B's own risk, $0.10 \times 56.3\%$	5.6%	5.6%
Privates in total, excluding C (root sum of squares)	6.5%	23.3%

In Setting 2, the privates' risk is roughly half theme, which the public book can hedge, and half A's own risk, which it cannot.

C is not a volatility. Describe it by its outcomes instead: −5% of NAV with probability 70%, +35% of NAV with probability 30%. Variance-based tools misstate such payoffs, so C is handled with exact scenario math in Step 8.

Why marks understate risk. A simple model of mark smoothing is:

$$\tilde{r}_t = (1 - \phi)\, r_t + \phi\, \tilde{r}_{t-1}$$

Here $\tilde{r}_t$ is the reported return and r_t the economic return. The stationary variance of this AR(1) process gives the reported vol:

$$\sigma_{\text{reported}} = \sigma \sqrt{\frac{1 - \phi}{1 + \phi}}$$

With $\phi = 0.7$ the factor is 0.42, so a private with 45% economic vol reports about 19%. For Setting 2 with the alpha book at about 16% vol, economic fund vol is about 24% and reported fund vol about 12%.

Consequence. A risk model built on marks would let the public book double its risk while measured fund vol barely moves. Use comparables-based betas and idiosyncratic vols, and treat marks as a lagged, smoothed signal of economic value.

Step 2 — The frictionless benchmark: completion hedge plus alpha

If everything could be traded, the public book would do two separate jobs:

1. set the whole fund's theme exposure to its target;
2. size its uncorrelated alpha against total NAV.

This benchmark is the starting point; Steps 3–5 add the frictions.

Derivation. Let $\mathbf{u}$ be the vector of public exposures and $\mathbf{y}$ the private weights, both as fractions of NAV. Let $\boldsymbol{\Sigma}$ be the covariance matrix, split into public (U) and private (P) blocks. The fund's log growth is:

$$g(\mathbf{u}) = \mathbf{u}'\boldsymbol{\mu}_U + \mathbf{y}'\boldsymbol{\mu}_P - \tfrac{1}{2}\left(\mathbf{u}'\boldsymbol{\Sigma}_{UU}\mathbf{u} + 2\mathbf{u}'\boldsymbol{\Sigma}_{UP}\,\mathbf{y} + \mathbf{y}'\boldsymbol{\Sigma}_{PP}\,\mathbf{y}\right)$$

Setting the gradient with respect to $\mathbf{u}$ to zero gives full Kelly. With risk aversion $\gamma = 1/\kappa$, only the first term is scaled:

$$\mathbf{u}^\star = \kappa\, \boldsymbol{\Sigma}_{UU}^{-1}\boldsymbol{\mu}_U \;-\; \boldsymbol{\Sigma}_{UU}^{-1}\boldsymbol{\Sigma}_{UP}\,\mathbf{y}$$

The first term is the speculative demand. The second is the minimum-variance hedge of what the privates already hold, and it does not depend on κ.

Consequence 1: uncorrelated alpha is sized to total NAV. If an alpha position is uncorrelated with the privates ($\boldsymbol{\Sigma}_{UP} = \mathbf{0}$ for that position), its optimal size as a fraction of total NAV does not depend on the privates at all. Its vol, measured on public NAV, therefore scales as $1/(1 - \pi)$:

- Setting 1: $1/0.85 = 1.18\times$ the all-public level;
- Setting 2: $1/0.40 = 2.50\times$;

- ratio between the settings: 2.125.

These are upper bounds; Steps 3–4 show why the frictions cut them.

Consequence 2: the theme exposure is a completion trade. With one theme factor, the fund's target theme exposure and the public book's share of it are:

$$B^\star = \kappa \frac{\mu_F}{\sigma_F^2}, \qquad u_F^\star = B^\star - B_{\text{priv}}$$

Kelly fraction κ, theme premium μ_F	Fund target $B^\star$	Public theme exposure, Setting 1 ($B_{\text{priv}} = 0.093$)	Public theme exposure, Setting 2 ($B_{\text{priv}} = 0.498$)
$\frac{1}{2}$, 4%	0.16	+0.07	−0.34
$\frac{1}{2}$, 6% (base)	0.245	+0.15	−0.25
$\frac{1}{2}$, 8%	0.33	+0.23	−0.17
1, 6%	0.49	+0.40	−0.01

In Setting 1 the public book can be long the theme. In Setting 2 it should be flat to net short under any of these choices. A long-biased public book in Setting 2 doubles down on the privates.

What a 45% position in A implies. Under fractional Kelly, holding weight w in an asset's idiosyncratic risk is optimal only if its alpha is:

$$\alpha = \frac{w\,\sigma_\varepsilon^2}{\kappa}$$

For A: $0.45 \times 0.1033 = 4.6\%$ a year at full Kelly, or 9.3% at half Kelly. So a 45% A is consistent with half Kelly only if you expect it to beat its theme-implied 5.4% by about 9 points a year. Run this check on any concentrated private.

What the benchmark assumes. It assumes the privates can be rebalanced, borrowed against, and trade at their economic value. None of that holds, which is what the next three steps address.

Step 3 — What illiquidity does to the denominator

A locked private still counts toward the public book's risk capacity, but only at a downside-weighted value. Under log utility alone, Setting 1 credits about 70–80% of private NAV and Setting 2 about 55–75%.

Model. The privates are locked until T and then worth a random amount X. The liquid sleeve is L. The fund's value as a function of liquid wealth is:

$$J(L) = \mathbb{E}\left[\log(L + X)\right]$$

Now add a small public bet εz that is uncorrelated with X, where each unit of z has mean μ and variance σ^2. Let $W = L + X$. Expanding to second order:

$$\mathbb{E}\left[\log(W + \varepsilon z)\right] \approx J(L) + \varepsilon\mu\,\mathbb{E}[W^{-1}] - \tfrac{1}{2}\varepsilon^2\sigma^2\,\mathbb{E}[W^{-2}]$$

Maximizing over ε gives Kelly sizing, with wealth replaced by an effective denominator D:

$$\varepsilon^\star = \frac{\mu}{\sigma^2} D, \qquad D = \frac{\mathbb{E}[W^{-1}]}{\mathbb{E}[W^{-2}]} = -\frac{J'(L)}{J''(L)}$$

D is the fund's risk tolerance toward liquid bets. Two summary numbers follow from it:

- the **credit** $\theta = (D - L)/P$, the share of private NAV that counts;
- the **multiplier** D/L, the public book's vol relative to an all-public fund.

Properties of D.

- If X is certain, $D = L + X$: full credit.
- If X is risky, $D < L + \mathbb{E}[X]$: the average is weighted toward states where the fund is poor.
- If W is lognormal with log-variance s^2, then $\mathbb{E}[W^{-1}] = e^{-m+s^2/2}$ and $\mathbb{E}[W^{-2}] = e^{-2m+2s^2}$, so:

$$D = \mathbb{E}[W]\, e^{-2s^2}$$

The log-variance s^2 grows with private volatility, private weight and T. Longer lockups and bigger, riskier positions therefore get less credit.

Worked results. These come from a Monte Carlo with 2 million paths. The privates are valued at T, with C fair-valued so that $\mathbb{E}[X] = P$. "Half hedged" means the public book continuously hedges half of the privates' theme beta.

Setting	T (years)	Credit θ	Multiplier D/L	Full credit, $1/(1-\pi)$
1	3 (base)	0.76	1.13	1.18
1	5 (stress)	0.68	1.12	1.18
2, unhedged	2.5 (base)	0.66	1.99	2.50
2, unhedged	4 (stress)	0.55	1.83	2.50
2, half hedged	2.5 (base)	0.76	2.15	2.50
2, half hedged	4 (stress)	0.67	2.00	2.50

Taken one at a time at their Setting 2 weights and $T = 2.5$, the credits are: A at 45%, 0.69; B at 10%, 0.79; C at 5%, 0.79.

Readings.

- The credit is set by downside dispersion, not by illiquidity as such. Hedging the theme raises it from 0.66 to 0.76.
- Duration matters: going from 2.5 to 4 years costs about 0.1 of credit.
- Risk aversion matters. With $\gamma = 3$ instead of log utility, the Setting 2 credit at 2.5 years falls from 0.66 to 0.48. Use a higher γ if your fractional Kelly reflects real aversion to drawdowns rather than estimation error.
- The pure cost of being unable to rebalance is small. If A drifts from 45% to 52% of NAV, the log-growth loss is $\frac{1}{2}\sigma_\varepsilon^2(\Delta w)^2 \approx 0.03\%$ a year. What matters is the level of concentration, not the drift.

Caveat. This is a one-period approximation. Rebalancing the public book as private values move narrows the gap a little; the path test in Step 5 catches what it misses.

Step 4 — The commercial cost of a high private share

Log utility attaches no penalty to crossing 50% private. If crossing it is costly for continuing investors, that cost has to be added explicitly. Doing so cuts Setting 2's alpha multiplier from about 2.0× to 1.2–1.7×.

What the cost is.

- forced use of liquidating capital accounts;
- valuation disputes;
- discounts if you have to sell;
- failing to meet normal cash redemptions;
- harder fundraising.

Count only what hits NAV per share of continuing investors. Franchise costs to the manager (AUM, fees) are real, but they are a separate decision; say explicitly whether you include them. Do not count lost trading capacity here either: that is the capital term in Step 6.

Model. The cost is $f(\pi)$ per year, quadratic above a 30% threshold and scaled so that $f(0.6) = f_0$:

$$f(\pi) = f_0 \left(\frac{(\pi - 0.3)^+}{0.3} \right)^2$$

The cost lasts τ years, until the private share falls back below 30%; in Setting 2, that is when A is sold. The total (stock) cost is $\Phi(\pi) = \tau \times f(\pi)$, and the objective becomes $U = \log N - \Phi(\pi)$.

Derivation. Let the public book earn x, a fraction of NAV. The private share moves to $\pi/(1+x) \approx \pi(1 - x + x^2)$. Expanding U to second order:

$$\mathbb{E}[\Delta U] \approx (1 + \pi\Phi')\,\mathbb{E}[x] - \tfrac{1}{2}\left(1 + 2\pi\Phi' + \pi^2\Phi''\right)\mathrm{Var}[x]$$

The optimal alpha size is therefore multiplied by:

$$\boxed{k(\pi) = \frac{1 + \pi\Phi'}{1 + 2\pi\Phi' + \pi^2\Phi''}}$$

The intuition: a public loss hurts twice, because it lowers NAV and raises π; a gain helps twice. The variance term grows faster than the mean term, so $k < 1$ whenever the cost is convex.

At $\pi = 0.6$, $\Phi' = 6.67\tau f_0$ and $\Phi'' = 22.2\tau f_0$. Writing $c = \tau f_0$, this simplifies to $k = (1 + 4c)/(1 + 16c)$.

Private share π	$f_0 = 1\%$, $\tau = 2.5$	$f_0 = 1\%$, $\tau = 4$	$f_0 = 3\%$, $\tau = 2.5$	$f_0 = 3\%$, $\tau = 4$
30% or less	1.00	1.00	1.00	1.00
40%	0.90	0.85	0.76	0.68
50%	0.84	0.78	0.67	0.58
60%	0.79	0.71	0.59	0.51

Combined multiplier. Multiply the multiplier from Step 3 by k. Base uses $T = \tau = 2.5$ years; stress uses 4.

Public alpha vol relative to an all-public fund	No commercial cost	$f_0 = 1\%$	$f_0 = 3\%$
Setting 1 (any case)	1.13	1.13	1.13
Setting 2, unhedged, base	1.99	1.56	1.18
Setting 2, unhedged, stress	1.83	1.29	0.93
Setting 2, half hedged, base	2.15	1.69	1.27
Setting 2, half hedged, stress	2.00	1.41	1.01

The answer for Setting 2. Hedged, in the base case, the alpha book can run 1.3–2.2× the all-public vol, with a central value of 1.7× at $f_0 = 1\%$. The implied credits are $\theta = 0.77$, 0.46 and 0.18 for $f_0 = 0$, 1% and 3%. If the exit slips, the range is 1.0–2.0×.

Relative to Setting 1, that is 1.1–1.9× (central 1.5×), against the mechanical 2.125×. It only matters if the alpha book is limited by risk rather than by capacity; if good ideas run out first, no multiplier helps.

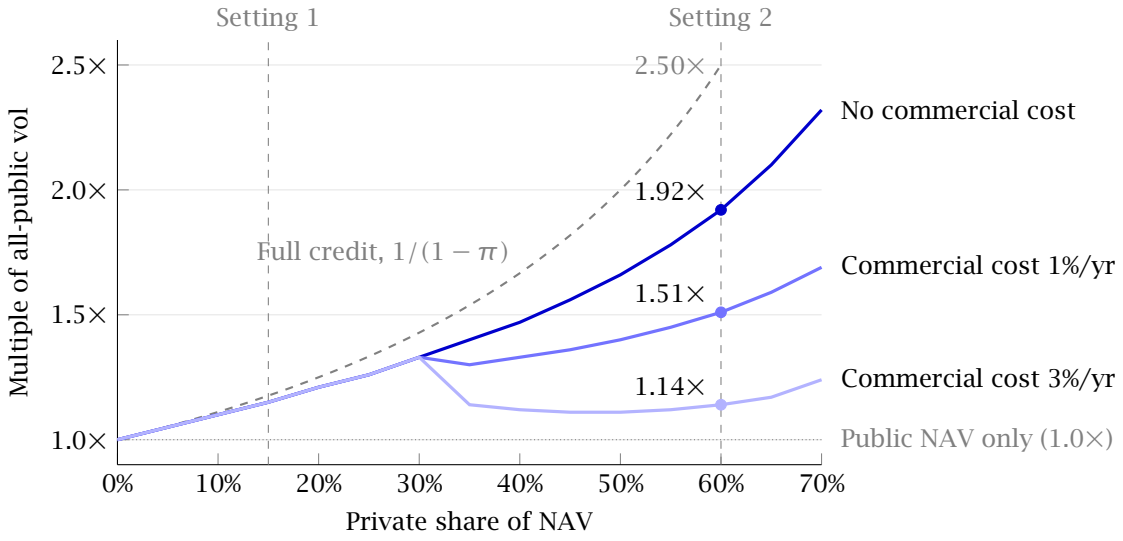


Figure 1: At 60% private, the public book can run 1.1–1.9× the all-public alpha vol, not 2.5×. Alpha vol the public book can run, as a multiple of what an all-public fund would run. Model from Steps 3–4: one unhedged private book with 45% vol, locked 2.5 years; commercial cost starts at 30% private, $\tau = 2.5$ years. Monte Carlo, 2 million paths.

Across private shares the pattern is the same. Below 30%, only the locked-privates effect operates, and it tracks full credit closely. Above 30%, the commercial cost flattens the curve, and at $f_0 = 3\%$ the public book gains almost no room as the private share rises. The kink at 30% is where the assumed penalty switches on; a smoother penalty would round it off without changing the levels at 50–60%.

Step 5 — Liquidity as a path constraint

In Setting 2 the binding constraint is cash in a rally, not losses in a crash. A futures or swap hedge of the privates, plus normal outflows, can consume most of a 40% liquid sleeve.

Sources-and-uses test. For each scenario and horizon, track cash through one reconciled calculation:

$$L_{\text{after}} = L + \text{public P\&L} + \text{hedge P\&L} - \text{cash redemptions} - \text{follow-ons} - \text{expenses}$$
$$\text{free equity} = L_{\text{after}} - \text{hedge margin} - \text{reserve} \;\geq\; \text{what the alpha book needs}$$

The reserve is the next 12 months of normal outflows: 5% redemptions, 1% expenses and 2% follow-ons, or 8% of NAV. Hedge margin is 20% of short notional.

The denominator effect. Paying a redemption q (a fraction of NAV) in cash raises the private share to $\pi/(1-q)$. A public loss ℓ does the same: $\pi/(1-\ell)$. Paying redemptions pro rata (cash for the liquid share, a liquidating capital account for the private share) leaves π unchanged.

Event	Setting 1: π after	Setting 2: π after
Start	15.0%	60.0%
10% cash redemption	16.7%	66.7%
20% cash redemption	18.8%	75.0%
Public loss of 10 points plus 5-point cash redemption	17.6%	70.6%
Any redemption paid pro rata	15.0%	60.0%

Stale marks also transfer value between investors. Redeemers paid in cash at marks above economic value take that difference from the investors who stay. In the Setting 2 crash below, marks exceed economic value by 4.9% of NAV. A 10% cash redemption therefore moves about 0.5% of NAV from stayers to leavers, which is itself a reason to redeem early.

Two one-year scenarios.

- **Crash.**
  - Market: theme -35%; public alpha -1.5σ.
  - Marks lag economic value: A -20% marked vs -31.5% economic; B -25% vs -32.5%; C -20% vs 0.
  - Cash flows: 10% redemptions in cash, 2% follow-ons (bridge rounds), 1% expenses.
- **Rally.**
  - Market: theme $+50\%$; public alpha flat.
  - Marks lag economic value: A $+30\%$ marked vs $+45\%$ economic; B $+30\%$ vs $+46\%$.
  - Cash flows: 5% redemptions, 3% follow-ons, 1% expenses.

In the table, "hedged" means short 0.25 of NAV of a theme basket (Step 2's target). Alpha vol is in % of public NAV. Free equity is liquid capital after hedge margin and reserve, as a fraction of starting NAV; it starts at 0.85 in Setting 1 and 0.40 in Setting 2.

Policy	Crash: reported / economic return	Crash: π	Crash: free equity	Rally: reported / economic return	Rally: π	Rally: free equity
S1, alpha 11.3%	−18.9% / −18.7%	19.0%	0.52	+2.0% / +3.6%	21.6%	0.68
S1, plus 0.15 theme long	−24.2% / −23.9%	20.5%	0.47	+9.5% / +11.1%	20.1%	0.75
S2 unhedged, alpha 10%	−19.5% / −24.4%	70.2%	0.15	+15.5% / +23.9%	71.9%	0.22
S2 unhedged, alpha 17%	−23.7% / −28.6%	74.7%	0.12	+15.5% / +23.9%	71.9%	0.22
S2 hedged, alpha 10%	−10.8% / −15.7%	62.5%	0.20	+3.0% / +11.4%	81.1%	0.03
S2 hedged, alpha 17%	−15.0% / −19.9%	66.0%	0.16	+3.0% / +11.4%	81.1%	0.03

Readings.

1. **In a crash, the hedge matters more than the alpha vol.** Hedged at 17% alpha vol, the fund keeps more free equity (0.16) and a lower π (66%) than unhedged at 10% (0.15, 70%).

2. **Marks hide economic moves in both directions.** In Setting 2, reported returns understate the crash by about 5 points and the rally by about 8. The hedged fund reports +3% in a +50% theme rally. Hedging trades reported vol for economic vol, and investors need to be told that in advance.

3. **In a rally, the linear hedge drains the sleeve.** It costs 12.5% of NAV in cash plus 7.5% in margin, leaving 0.03 of free equity. The largest short that still leaves 0.10 of NAV for the alpha book is 0.16 of NAV; for 0.15 it is 0.09. That is well under the 0.25 target.

Ways out.

- **Hedge with options.** A 1-year put on 0.25 of NAV of the basket (35% vol) costs about 2.2% of NAV at a 90% strike, 1.2% at 80%, or 1.6% for a 90/70 put spread. There is no further cash drain in a rally.
- **Hedge less** with futures or swaps.
- **Shrink A** through a secondary sale.
- **Raise capital.**

Rally states are also where A lists and fundraising works, but the path test should not count on either.

After A lists. Standard lockup agreements usually bar hedging the locked shares directly. Proxy baskets are usually allowed; check the agreement.

Fundraising. Raising new cash f (a fraction of NAV) dilutes the private share to $\pi/(1+f)$. From 60%:

- to 50% needs a raise of 20% of NAV;
- to 30% needs 100%;
- to 25% needs 140%.

Fundraising is harder after losses and when π is high, so give it no credit in stress. Price subscriptions so that new money does not buy stale marks.

Step 6 — The hurdle for a marginal private

A private must beat its theme-implied return by an alpha that pays for four costs. If the public book re-hedges the private's beta, the theme risk itself costs nothing at the margin.

Exact test. Move a fraction δ of NAV out of a displaced asset a (gross payoff X_a) and into private j (gross payoff X_j), over a common horizon. Let G be the gross outcome of the rest of the fund. Then:

$$\Delta U = \mathbb{E}\left[\log\left(1+\frac{\delta\,(X_j - X_a)}{G}\right)\right], \qquad \left.\frac{dU}{d\delta}\right|_{\delta=0} = \mathbb{E}\left[\frac{X_j - X_a}{G}\right]$$

$1/G$ is the log investor's state price: a dollar counts more in states where the fund is poor. Use this form directly for skewed or binary payoffs such as C.

Approximation for smoother payoffs. Using $g \approx \mu - \frac{1}{2}\sigma^2$ and an annual cost c_j:

$$\Delta g \approx \delta\,[\mu_j - \mu_a - \mathrm{Cov}(r_j - r_a,\, r_W)] - \tfrac{1}{2}\delta^2\,\mathrm{Var}(r_j - r_a) - \delta\, c_j$$

Setting $\Delta g \geq 0$ gives the hurdle:

$$\mu_j - \mu_a \;\geq\; \mathrm{Cov}(r_j - r_a,\, r_W) + \tfrac{\delta}{2}\,\mathrm{Var}(r_j - r_a) + c_j$$

Apply it with a completion hedge. Write $r_j = \beta_j F + \varepsilon_j + \alpha_j$. Once the public book re-hedges, the displaced package is capital from the funding source (return r_K) plus β_j units of theme exposure. The theme terms cancel, and the hurdle becomes an alpha requirement:

$$\alpha_j \;\geq\; \underbrace{r_K(1 \pm \beta_j m) - \mathrm{Cov}(r_K, r_W)}_{\text{capital}} + \underbrace{w_j\,\sigma^2_{\varepsilon j}}_{\text{concentration}} + \underbrace{\tfrac{\delta}{2}\left(\sigma^2_{\varepsilon j} + \sigma^2_K\right)}_{\text{size}} + \underbrace{c_j(\pi)}_{\text{commercial liquidity}}$$

The four components.

1. **Capital (the leverage constraint).** A private has zero collateral value, so every $1 in it withdraws $1 of equity from the public book.
   - **Capital-bound** public book (at its gross or margin limit): r_K is its return on equity, 12% here, less its own covariance with the fund (about 1%).
   - **Margin on the re-hedge:** $\beta_j \times m$. It adds when the hedge means more shorts (Setting 2) and subtracts when it replaces public longs (Setting 1).
   - **Capacity-bound** public book (good ideas, not capital, are scarce): $r_K \approx 0$.
2. **Concentration.** $w_j\sigma^2_{\varepsilon j}$ is the fund's existing weight in the private's own risk, times that risk's variance. This is where the private share bites: it is 0 for a first A in Setting 1 and 4.65% for more A in Setting 2.
3. **Size.** $(\delta/2)$ times the variance of the swap. It is small for a 5% ticket.
4. **Commercial liquidity.** $c_j = f'(\pi) \times$ the fraction of the private's life spent with π above 30%, where $f'(\pi) = 2f_0(\pi - 0.3)/0.09$. It is zero at 15% private and $6.7f_0$ at 60%: 6.7% or 20% a year for $f_0 = 1\%$ or 3%. The marginal cost is far above the average cost because the penalty is convex.

Without re-hedging. Add $\beta_j(B\sigma^2_F - \mu_F)$, where B is the fund's theme dollar-beta. In Setting 2 unhedged this is about 0 under full-Kelly log utility. At the half-Kelly target it is -2.7% under pure log utility, which would want more theme exposure than half Kelly. Under the half-Kelly policy itself ($\gamma = 2$, so the covariance counts twice) it is zero by construction.

Expected return vs IRR. These hurdles are expected (probability-weighted) annual returns. For skewed payoffs the expectation is above the base case; for a lognormal asset the median compound return sits $\frac{1}{2}\sigma^2$ below the mean, which is 10 points for A. Underwrite on scenario-weighted expected multiples, not base-case IRRs.

Step 7 — A, B and C compared

Per dollar, B is the most dangerous for a fund with a liquidity constraint. A is dangerous only through its size. C is cheap if it is small and spread across several names.

	A: near-IPO mega-cap	B: late stage	C: early stage
Risk character	Large-cap tech equity, beta 0.9 to the theme	Theme beta plus financing and dilution risk	Binary: 70% total loss, 30% 8×
Hedgeable with publics	Mostly (correlation about 0.7)	Partly (about 0.5)	Barely
Time to usable liquidity, base / stress	2.5y / 4y (IPO, lockup, block sell-down)	3y / 5y	4y or more either way
Bad state	IPO window shuts; marks catch down late	Exit slips, down round, bridge financing needed	Fails: goes to 0 and leaves π
Good state	Lists; the sell-down lowers π	Lists; π falls	8×: becomes 35% of original NAV and raises π; secondaries usually open
Main risk to manage	Concentration; repricing at IPO against the mark	Liquidity date slips in exactly the bad states	Skew, follow-ons, correlation with the same boom
How to size	Idiosyncratic budget (Step 2 check); hedge its beta	Smaller than its expected return alone suggests	Many small tickets; size on committed capital including follow-ons

A: "likely to succeed" is not "low risk at this price."

- **Size:** a 30% fall in A's value at a 45% weight costs 13.5% of NAV.
- **Deal terms:** preferred terms such as IPO ratchets and liquidation preferences cushion the downside relative to common stock, so put them in the scenarios.
- **The IPO itself:** at listing, the mark on 45% of NAV jumps to the market price. Listing changes the reported price well before it changes usable liquidity.

B: wrong-way liquidity. Its time to liquidity lengthens exactly when the fund wants cash, and the fund may have to fund bridge rounds at the same moment. Only part of its risk can be hedged. That is why B's hurdle in Step 8 is close to A's even at a much smaller weight.

C: the cheapest risk, if it is small.

- **If it fails,** the position leaves the book: π falls, but only because wealth was destroyed.
- **If it succeeds,** π rises in a state where the fund is rich and secondary markets are open, which is a good problem.
- **What actually costs:** single-name skew, follow-on commitments, and correlation with the same boom that drives A and B. With a correlation of 0.5 to the theme, C's required success multiple rises by about 10% (Step 8).

Step 8 — Worked hurdles

At 15% private, A needs its theme-implied return plus 0–9 points and B plus 4–13. At 60% private, both need roughly 5–30 more points, driven by concentration and the commercial cost of illiquidity.

A and B. The table shows the required alpha, per year, over the theme-implied return (A 5.4%, B 5.6%). It assumes a 5% ticket and a public book that re-hedges the new beta. In Setting 1 the public book is long the theme at its target; in Setting 2 it is short.

Private, setting	Capital	Concentration	Size	Commercial (f_0 1% / 3%)	Required alpha, capital-bound	Required alpha, capacity-bound
A at 15% private	8.8%	0	0.3%	0	9.0%	0.3%
B at 15% private	8.7%	3.2%	0.8%	0	12.7%	4.0%
A at 60% private	13.0%	4.7%	0.3%	6.7% / 20.0%	24.6% / 38.0%	11.6% / 24.9%
B at 60% private	13.1%	3.2%	0.9%	5.6% / 16.7%	22.7% / 33.8%	9.5% / 20.6%

The total expected return required is the theme-implied return plus the alpha. For example, A at 15% private in a capital-bound fund needs 14.4%; A at 60% private in a capacity-bound fund with $f_0 = 1\%$ needs 17.0%.

How the components are computed.

- **Capital, A at 15%:** $12\% \times (1 - 0.9 \times 0.2) - 0.85 \times 11.3\%^2 = 9.84\% - 1.09\% = 8.75\%$.
- **Capital, A at 60%:** $12\% \times (1 + 0.18) - 0.40 \times 17\%^2 = 14.2\% - 1.2\% = 13.0\%$.
- **Concentration:** A at 60%, $0.45 \times 0.1033 = 4.65\%$; B in both settings, $0.10 \times 0.317 = 3.2\%$.
- **Size, A:** $0.025 \times (0.1033 + 0.0128) = 0.29\%$.
- **Commercial, A at 60%:** $6.7 f_0 \times 1.0$, because A's whole life falls in the high-π window.
- **Commercial, B at 60%:** $6.7 f_0 \times 2.5/3$, because B stays locked about six months after A is sold.

Readings.

- The largest single swing is whether the public book is capital-bound: 9–13 points.
- Going from 15% to 60% private adds concentration (+4.7 for A) and the commercial cost (+6 to +20).
- **Should you sell some A?** In Setting 2, suppose you believe A's alpha is 9.3% (the half-Kelly value implied by holding 45%, Step 2). The capacity-bound hurdle at $f_0 = 1\%$ is 11.6%. Each dollar of A then falls short by 2.3 points a year for about 2.5 years. So a secondary sale at a discount below about 6% adds value. With $f_0 = 3\%$, or a capital-bound public book, the break-even discount is much larger.

C, with exact log utility. C succeeds with probability 30% and pays M times its cost after 4 years; otherwise it pays zero. The table gives the break-even $M^\star$, from the exact test in Step 6, with the implied expected-value IRR in brackets:

$$\text{EV-IRR} = (0.3\,M^\star)^{1/4} - 1$$

The rest of the fund G is simulated with its own risks, including the C already held. The stipulated C has $M = 8$, an EV-IRR of 24.5%.

Setting, displaced asset, C's correlation with the theme	Tiny position	5% in one name	5% over five names	10% in one name (with follow-ons)
S1, cash, 0	3.33× (0.0%)	3.47× (1.0%)	3.36× (0.2%)	3.63× (2.2%)
S1, cash, 0.5	3.67× (2.4%)	3.84× (3.6%)	3.72× (2.8%)	4.03× (4.9%)
S1, alpha capital, 0	5.14× (11.5%)	5.50× (13.3%)	5.22× (11.9%)	5.91× (15.4%)
S1, alpha capital, 0.5	5.66× (14.2%)	6.09× (16.3%)	5.80× (14.8%)	6.59× (18.6%)
S2, cash, 0.5	3.72× (2.8%)	3.96× (4.4%)	3.79× (3.3%)	4.25× (6.2%)
S2, alpha capital, 0.5	5.63× (14.0%)	6.23× (16.9%)	5.82× (15.0%)	6.96× (20.2%)

In Setting 2, add the commercial cost to the EV-IRR hurdle: $6.7 f_0 \times 2.5/4$, which is +4.2 points at $f_0 = 1\%$ and +12.5 at 3%.

Readings.

- **A tiny independent C is risk-neutral at the margin.** It needs only an expected multiple equal to what it displaces: 3.33× on success is 1.0× expected against cash. Under log utility, independent risk costs nothing at the margin.
- **Size, correlation and follow-ons each add 1–3 points** of EV-IRR. Spreading 5% over five names saves about 1.5 points.
- **The IRR if it works is misleading.** A 6.1× break-even is a 57% IRR if C succeeds, but only a 16% EV-IRR.
- **The stipulated C clears almost everywhere.** The one exception is Setting 2 with a capital-bound public book and $f_0 = 3\%$, where the hurdle is 16.9% + 12.5% = 29.4%.

Take-aways

1. **One objective, two constraints.** Measure risk and value on total economic NAV per share. Treat illiquidity as two things: a smaller effective denominator for alpha, and a cash constraint along adverse paths.
2. **The public book is first a completion portfolio.** Its theme exposure is the fund's target minus what the privates hold. At 15% private it can be long; at 60% it should be flat to short.
3. **Private NAV counts at a downside-weighted value.** The credit $\theta = (D - L)/P$, with $D = \mathbb{E}[W^{-1}]/\mathbb{E}[W^{-2}]$, falls with the private share, private volatility and time to usable liquidity. Hedging the privates' theme beta raises it.
4. **The commercial cost of a high private share is separate and convex.** Decide explicitly whether it covers only continuing investors or also the manager's franchise.

5. **The numbers.** In Setting 1, manage to total NAV (about 1.13× the all-public vol). In Setting 2, the alpha book runs about 1.3–2.2× (central 1.7×), not 2.5×. The ratio to Setting 1 is about 1.5×, not 2.125×.

6. **In Setting 2 the binding constraint is cash in a rally.** A 0.25-of-NAV futures or swap hedge plus normal outflows can exhaust the liquid sleeve. Hedge with options, hedge less, or shrink A.

7. **Marks understate risk and lag in both directions.** At 60% private they show about half the true vol. Hedging trades reported vol for economic vol.

8. **A private's hurdle** is its theme-implied return plus capital, concentration, size and commercial costs. Whether the public book is capital-bound is the largest single input.

9. **B is the most dangerous per dollar** because its liquidity is wrong-way. A is dangerous through its size. C is dangerous through skew and follow-ons, so diversify it.

10. **Pay redemptions pro rata above a threshold.** This keeps π constant and removes the incentive to redeem at stale marks.

11. **Fundraising dilutes π only slowly and is procyclical.** Going from 60% to 30% needs a raise equal to current NAV, so give it no credit in stress.

Inputs to set

The conclusions are robust in direction; the magnitudes depend mostly on these inputs, listed roughly in order of impact.

Input	Used in	Value used here	Why it matters
Is the public book capital-bound or capacity-bound, and what is its marginal return on equity?	Step 6	12% or 0	Largest single swing in every hurdle (9–13 points)
f_0: annual cost to continuing investors of sitting at 60% private	Steps 4, 6	1% and 3%	Sets how fast the multiplier falls and hurdles rise above 30% private
A's time to usable liquidity, base and stress	Steps 3–5	2.5y / 4y	Sets A's credit and how long the commercial cost lasts
Theme premium and Kelly fraction	Step 2	6%, $\frac{1}{2}$	Sets the target theme exposure, and so the hedge
Private betas and idiosyncratic vols, from comparables	Steps 1–3	Setup table	Drive the concentration charges and the credit
Risk aversion behind the Kelly fraction: estimation error or real drawdown aversion	Step 3	log ($\gamma = 1$)	$\gamma = 3$ cuts the Setting 2 credit from 0.66 to 0.48
C's success probability, multiple, correlation with the theme, follow-on policy	Step 8	30%, 8×, 0–0.5	Decide C's size and number of names
Threshold for paying redemptions pro rata	Step 5	Not set	Caps the denominator effect and the stale-mark transfer
Hedge instrument: futures or swaps vs puts	Step 5	Short basket; puts as the alternative	Sets rally-state cash needs